TWISTED COHOMOLOGY AND TWISTED FIELD THEORIES

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The goal is to prove the following theorem. The main result is in relating field theories and super geometry to other well known models of ∞-local systems.

Theorem 0.0.1. *Let M be a manifold. The following stable ∞-categories are equivalent:*

1. *The category* $\mathrm{Fun}_{\mathcal{S}\mathrm{Mfld}}(0|1 - \mathsf{Bord}(M), \mathrm{sVect})$ *of* $0|1 - \mathrm{EFT}$ *twists over* M.
2. *The category* $\mathsf{sVect}^{\mathbb{R}^{0|1}}(\Pi TM)$ *of* $\mathbb{R}^{0|1}$*-equivariant super vector bundles on* ΠTM.
3. *The category of* $\mathsf{Flat}_{\mathbb{Z}/2}(M)$ *of super vector bundles with flat superconnection on* M .
4. *The category* $\mathrm{Fun}_{\infty}(\Pi_{\infty}(M), \mathcal{M}\mathrm{od}(H\mathbb{R}[u, u^{-1}]))$ *of twists for* 2*- periodic de Rham cohomology.* $(|u| = 2)$

There is a non-periodic version. Its setup requires us to use the category $\mathrm{c}\mathcal{S}\mathrm{Mfld}$ of complex supermanifolds. We will explain why this result does not hold over $\mathcal{S}\mathrm{Mfld}$.

Theorem 0.0.2. *Let M be a manifold. The following stable ∞-categories are equivalent:*

1. *The category* $\mathrm{Fun}_{\mathrm{c}\mathcal{S}\mathrm{Mfld}}(0|1 - \mathsf{Bord}(M), \mathrm{sVect})$ *of* $0|1 - \mathrm{TFT}$ *twists over* M.
2. *The category* $\mathsf{sVect}_{\mathbb{C}}^{\mathbb{R}^{0|1}_{\mathrm{cs}}}(\Pi TM)$ *of* $\mathrm{End}(\mathbb{R}^{0|1}_{\mathrm{cs}})$*-equivariant complex super vector bundles on* $\Pi_{\mathbb{C}}TM$.
3. *The category* $\mathsf{Flat}_{\mathbb{Z}}(M)$ *of* $\mathbb{Z}$*-graded vector bundles with flat* $\mathbb{Z}$*-graded connection on* M .
4. *The category* $\mathrm{Fun}_{\infty}(\Pi_{\infty}(M), \mathcal{M}\mathrm{od}(H\mathbb{C}))$ *of twists of complex de Rham cohomology.*
5. *The category* $\mathcal{S}\mathrm{hv}_{\infty}^{l.c.}(M, \mathcal{D}(\mathbb{C}))$ *of ∞-local systems with values in the derived category of* $\mathbb{C}$*. This consists of homotopy sheaves that are locally constant.*

In particular, we will show how these categories are to be regarded as ∞-categories.

1. 0|1 Field Theories

We now provide a rapid summary of [HKST11]. We start with the necessary prerequisites on supermanifolds. The internal hom $\underline{\mathrm{Hom}}(M, N)$ can be defined on any presheaf category as the presheaf $S \mapsto \mathrm{Hom}(M \times S, N)$ and $\underline{\mathrm{Hom}}(M, -)$ is the right adjoint of $- \times M$. We work in the category $\mathcal{S}\mathrm{Mfld}$ of supermanifolds and generalized supermanifolds $\mathcal{P}\mathrm{re}\mathcal{S}\mathrm{hv}(\mathcal{S}\mathrm{Mfld})$. For any supermanifold M the presheaf $\underline{\mathrm{Hom}}(\mathbb{R}^{0|1}, M)$ is representable by the shifted tangent bundle ΠTM. ΠTM is the supermanifold (M, Ω^*_M) whose sheaf of functions are the differential forms.

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Similarly, it is possible to determine $\mathrm{End}(\mathbb{R}^{0|1}) = \underline{\mathrm{Hom}}(\mathbb{R}^{0|1}, \mathbb{R}^{0|1}) \cong \mathbb{R} \ltimes \mathbb{R}^{0|1}$ acting on $\mathbb{R}^{0|1}$ by dilation and translation:

$$(t, \eta).\theta = t\theta + \eta\,.$$

The invertible self maps form the diffeomorphism group $\mathrm{Diff}(\mathbb{R}^{0|1}) = \mathbb{R}^\times \ltimes \mathbb{R}^{0|1}$. The other interesting subgroup is the group of isometries $\mathrm{Iso}(\mathbb{R}^{0|1}) := \{\pm 1\} \ltimes \mathbb{R}^{0|1}$ preserving $d\theta^2$. This induces an $\mathbb{R}^{0|1}$-action on $\underline{\mathrm{Hom}}(\mathbb{R}^{0|1}, M) \cong \Pi TM$ that induces the de Rham differential $\theta.\omega = \omega + \theta d\omega$.

The first group defines the geometry of (all) $0|1$ bordisms and the second one defines the Euclidean bordism category. We will see how below.

We need to work in families over $\mathcal{S}\mathrm{Mfld}$. There are several reasons for this. Firstly, it incorporates smooth dependence of the parameters. Secondly, working with supermanifolds only makes sense in families: otherwise there are no maps $\mathbb{R}^{0|1} \to M$ for M an ordinary manifold. On the contrary, there are many interesting maps $S \times \mathbb{R}^{0|1} \to M$ represented by the odd tangent bundle.

We now only present the product bordisms in $0|1 - \mathsf{Bord}$ and $0|1 - \mathsf{EBord}$. General connected bordisms are locally given by products. For both cases, a product family of bordisms parametrized by the supermanifold S is given by $S \times \mathbb{R}^{0|1}$. The morphisms in both cases are bundle maps

$$\begin{array}{ccc} S \times \mathbb{R}^{0|1} & \longrightarrow & S \times \mathbb{R}^{0|1} \\ & \searrow \quad \swarrow & \\ & S & \end{array} \quad . \tag{1}$$

In the case of $0|1 - \mathsf{Bord}$ they need to be fibrewise diffeomorphisms, i.e. given by a map $S \to \mathrm{Diff}(\mathbb{R}^{0|1})$. In the case of $0|1 - \mathsf{EBord}$ they need to be fibrewise isometries, i.e. given by a map $S \to \mathrm{Iso}(\mathbb{R}^{0|1})$. This restricts the gluing maps. Note that the term "bordism" in dimension zero does not mean anything since the boundary is of dimension -1, i.e. empty by definition. The categories $0|1 - \mathsf{Bord}(M)$ and $0|1 - \mathsf{EBord}(M)$ are defined by requiring the bordisms to come equipped with maps to M and morphisms to also intertwine these.

The key result is that $0|1 - \mathsf{Bord}(M)$ is the free monoidal category on the stack $\Pi TM // \mathrm{Diff}(\mathbb{R}^{0|1})$ and $0|1 - \mathsf{EBord}$ is the free monoidal category on the stack $\Pi TM // \mathrm{Iso}(\mathbb{R}^{0|1})$.

Theorem 1.0.1. *We can classify $0|1$-TFTs and $0|1$-EFTs as invariant functions on ΠTM, i.e. closed forms:*

$$0|1 - \mathrm{TFT}\,(X) \cong \mathrm{Fun}(\Pi TM // \mathrm{Diff}(\mathbb{R}^{0|1}), \mathbb{R}) \cong \Omega^0(M)_{\mathrm{cl}}$$

$$0|1 - \mathrm{EFT}\,(X) \cong \mathrm{Fun}(\Pi TM // \mathrm{Iso}(\mathbb{R}^{0|1}), \mathbb{R}) \cong \Omega^{\mathrm{even}}(M)_{\mathrm{cl}}$$

The forms are closed since $\theta.\omega = \omega$ if and only if $d\omega = 0$.

To get higher degree (resp. odd) forms we need to introduce *twists*. The trivial twist is the trivial real line bundle over ΠTM. Untwisted field theories are (equivariant) sections of this trivial bundle or, equivalently, functions.

Definition 1.0.2. *A twist for $0|1$-TFTs is a vector bundle over $\Pi TM // \mathrm{Diff}(\mathbb{R}^{0|1})$, i.e. a fibred functor in*

$$0|1 - \mathrm{Twi}(M) := \mathrm{Fun}_{\mathcal{S}\mathrm{Mfld}}(\Pi TM // \mathrm{Diff}(\mathbb{R}^{0|1}), \mathrm{sVect})\,.$$

A twist for $0|1$*-EFTs is a vector bundle over* $\Pi TM//\operatorname{Iso}(\mathbb{R}^{0|1})$*, i.e. a fibred functor in*

$$0|1-\mathrm{ETwi}(M) := \mathrm{Fun}_{\mathrm{SMfld}}(\Pi TM//\operatorname{Iso}(\mathbb{R}^{0|1}), \mathrm{sVect})\,.$$

Question 1.0.3. *(1) Maybe we need a variant of EFT twists that extend smoothly to* $\mathbb{R}^\times \subset \mathbb{R}$*.*

(2) Maybe we need to also consider the oriented groups $\mathbb{R}^{0|1}$ *and* $\mathbb{R}_{>0} \ltimes \mathbb{R}^{0|1}$*.*

We will come back to this definition below when we define the dg-categories of twists.

2. Recollection of stable ∞-categories, dg-categories and A_∞-functors

Let $\mathcal{V}$ be a monoidal category. Recall that a *category enriched in* $\mathcal{V}$ satisfies the analogous axioms of a category except that the 'space' of arrows $\hom(x,y)$ is an object of $\mathcal{V}$. Moreover, composition is a morphism $\hom(y,z)\otimes\hom(x,y)\to\hom(x,z)$ satisfying the usual associativity and unitality conditions. If $F:\mathcal{V}\to\mathcal{U}$ is a lax monoidal functor (i.e. the natural transformation $F(x)\otimes F(y)\to F(x\otimes y)$ does not need to be an isomorphism and similarly for the unit), then any category enriched in $\mathcal{V}$ is also enriched in $\mathcal{U}$ by applying F to $\hom(x,y)$. A *dg-category* (over a ring k) is a category enriched in chain complexes Ch_k. The *homotopy category* $\mathrm{Ho}(\mathcal{C})$ *of a dg-category* $\mathcal{C}$ is the category whose 'spaces' of arrows are given by the homology $H_0\hom(x,y)$ of the chain complexes.

By the term ∞*-category* we refer to any of the equivalent models of ∞-categories such as quasicategories. In an ∞-category there is a space (or ∞-groupoid) of morphisms $\mathcal{M}\mathrm{ap}(x,y)$ [1].

The *dg-nerve* $N_{dg}(\mathcal{C})$ of a dg-category $\mathcal{C}$ is a quasicategory. Roughly speaking, the 1-morphisms of $N_{dg}(\mathcal{C})$ are the cycles in $\hom(x,y)_0$, the 2-morphisms are the witnesses that two cycles are homologous and the higher morphisms are higher coherences. We say $N_{dg}(\mathcal{C})$ is the ∞-category *presented by* the dg-category $\mathcal{C}$ and we sometimes abusively identify $\mathcal{C}$ and $N_{dg}(\mathcal{C})$.

More functorially, the dg-nerve N_{dg} is a right Quillen functor from dg-categories with the Dwyer-Kan model structure to simplicial sets with the Joyal model structure and hence preserves weak equivalences betweens fibrant objects [Lur17, 1.3.1.19]. All dg-categories are fibrant and the fibrant objects in the Joyal model structures are the quasicategories. Weak equivalences in the Dwyer-Kan model structure, so-called *quasiequivalences*, are the morphisms of dg-categories $f:\mathcal{C}\to\mathcal{D}$ satisfying:

(1) The map $F:\mathcal{C}(c,c')\to\mathcal{D}(F(c),F(c'))$ is a quasiisomorphism of chain complexes.
(2) The functor $\mathrm{Ho}(F):\mathrm{Ho}(\mathcal{C})\to\mathrm{Ho}(\mathcal{D})$ is an equivalence of (homotopy) categories. [2]

To summarize, quasiequivalences of dg-categories are mapped to (weak) equivalences of quasicategories. Phrasing this model independently, weakly equivalent dg-categories present equivalent ∞-categories.

[1] In fact, categories enriched in topological spaces are a model for ∞-categories.
[2] It is sufficient that $\mathrm{Ho}(F)$ is essentially surjective.

Dg-categories are a special case of A_∞-categories and dg-functors are special cases of A_∞-functors. However, by Theorem A in [COS19] two dg-categories are A_∞-quasiequivalent if and only if they are quasiequivalent in the dg sense (via a zig-zag). We record this here because we need this statement once.

An ∞-category is called *stable* if it has a zero object (both initial and terminal) and

(1) All morphisms admit kernels and cokernels.
(2) Fiber sequences and cofiber sequences agree, i.e., the following diagram is a pushout if and only if it is a pullback:

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & Z \end{array}$$

In a stable ∞-category the space of morphisms is canonically an infinite loop space, i.e. it is enriched in the (∞-)category Sp of spectra. The ∞-category of functors into a stable ∞-category is again stable.

Dg-categories are very much related to stable ∞-categories. While it is not true that the dg-nerve is always a stable ∞-category, it does come equipped with mapping spectra [**?**]. But, importantly, $N_{dg}(\mathrm{Ch}(R))$ is a stable ∞-category.

The stable ∞-category Sp is symmetric monoidal under the smash product. There is a lax monoidal functor $H : \mathcal{D}(\mathbb{Z}) \to \mathrm{Sp}$ from the derived ∞-category of abelian groups to spectra. The derived ∞-category of abelian groups is the ∞-categorical localisation $\mathrm{Ch}[qi^{-1}]$ of chain complexes at quasiisomorphisms. For any graded ring R, the cohomology $H^*(-;R)$ with values in R is represented by the ring spectrum $HR := H(R)$. Moreover, H induces an equivalence $\mathcal{D}(R) \simeq \mathrm{Ch}(R)[qi^{-1}] \simeq \mathcal{M}\mathrm{od}(HR)$ for any graded ring R. This is even a symmetric monoidal equivalence.

For example, for any ring R we can form the graded ring $R[u,u^{-1}]$ with a free generator of degree $|u| = 2$ and the corresponding spectrum $HR^{per} := HR[u,u^{-1}]$ classifies the periodic R-cohomology, in the sense that $H^*(-;R[u,u^{-1}]) \cong \bigoplus_{n\in\mathbb{Z}} H^{*+2n}(-;R)$. The module ∞-category $\mathcal{M}\mathrm{od}(HR[u,u^{-1}])$ is equivalent to the derived category $\mathcal{D}(R[u,u^{-1}])$. A chain complex over $R[u,u^{-1}]$ in turn is a chain complex C_* over R together with an isomorphism of chain complexes $u : C_* \cong C_{*+2}$. Hence, there is an equivalence of ∞-categories $\mathcal{M}\mathrm{od}(HR^{per}) \simeq \mathrm{Ch}^{\mathbb{Z}_2}(R)[qi^{-1}]$.

The objects of $\mathrm{Ch}^{\mathbb{Z}_2}(R)$ are $\mathbb{Z}_2$-graded R-modules equipped with odd differentials which can equivalently be regarded as 2-periodic chain complexes. The quasiisomorphisms are those inducing isomorphisms on homology in even and odd degree.

What kind of monoidal structure do I want? E.g. recall the one for Ch.

3. What are Twists?

We start with an account of what classically is called a twist.

(1) *Differential Geometry.* Differential Geometry computes the real cohomology of a space in terms of de Rham cohomology and differential forms. De Rham cohomology of a manifold M is the cohomology of the cochain complex

$$0 \longrightarrow C^\infty(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \dots \xrightarrow{d} \Omega^n(M) \longrightarrow 0 \ .$$

A *twist* for the de Rham cohomology of M is now given by a vector bundle $E \to M$ and a *flat* connection $\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E) = \Omega^1(M;E)$.

The map ∇ needs to satisfy the Leibniz rule $\nabla(f\sigma) = df\sigma + f\nabla\sigma$ for $f \in C^\infty(M)$ and $\sigma \in \Gamma(E)$. Recall that ∇ extends uniquely to a map $\Omega^*(M;E) \to \Omega^{*+1}(M;E)$ satisfying the (more general) Leibniz rule $\nabla(\omega\sigma) = d\omega\sigma + (-1)^{|\omega|}\omega\nabla\sigma$ for $\omega \in \Omega^*(M)$ and $\sigma \in \Omega^*(M;E)$. The composition $\nabla \circ \nabla$ is C^∞-linear and hence given by a bundle endomorphism $R_\nabla \in \Omega^2(M;\mathrm{End}(E))$ called *curvature*. The connection is flat if $R_\nabla = 0$ or equivalently if $\nabla^2 = 0$. Hence, the following is a cochain complex:

$$0 \longrightarrow \Gamma(E) \xrightarrow{\nabla} \Omega^1(M;E) \xrightarrow{\nabla} \Omega^2(M;E) \xrightarrow{\nabla} \dots \xrightarrow{\nabla} \Omega^n(M;E) \longrightarrow 0 \ .$$

By definition, this computes the twisted de Rham cohomology of M with respect to the twist (E,∇).

(2) *Topology.* The twists in topology are often referred to as *local systems*. We formulate it in terms of the fundamental groupoid $\Pi_1(M)$ of a topological space M. If M is connected, we can equivalently think of the fundamental group $\pi_1(M)$. A local system is then given by a representation of $\Pi_1(M)$, i.e. a functor $\Pi_1(M) \to \mathcal{V}\mathrm{ec}$. [3] Spelling this out for a connected manifold, this consists of the choice of a vector space V and a representation $L : \pi_1(M) \to \mathrm{GL}(V)$. The fundamental group π acts on the universal covering space $\widetilde{M}$ by Deck transformations. Hence, the singular simplices $C_*(\widetilde{M})$ also come with a π-action or, equivalently, with a $\mathbb{Z}\pi$ action. We can hence define the cohomology with respect to the local system L as the cohomology of the cochain complex of equivariant cochains $\mathrm{Hom}_{\mathbb{Z}\pi}(C_*(\widetilde{M});L)$.

(3) *Sheaf Theory.* The most concise definition is the following. A local system on a topological space M is a locally constant sheaf (of vector spaces).

A sheaf is constant if it is the sheafification of a constant presheaf that assigns to every open set the same vector space V. A sheaf $\mathcal{F}$ is locally constant if there is a cover (U_i) of M such that $\mathcal{F}|_{U_i}$ is a constant sheaf. The cohomology with respect to the local system $\mathcal{F}$ is just the sheaf cohomology $H^i(M;\mathcal{F})$, i.e. the cohomology of the right derived functor of global sections $R\Gamma^i(M;\mathcal{F})$.

Let us say a few words about the equivalence between (1), (2) and (3). Of course all spaces should be 'nice' enough. (1) $\to$ (2) is basically a local-global statement as in Lie theory. We make this precise below. State a theorem relating the fundamental groupoid and its representations. Also state the generalization to the path groupoid. The key input is that flat connections have a homotopy invariant parallel transport that gives the representation of the fundamental groupoid. For (2) $\to$ (3) we form the associated bundle $\widetilde{M} \times_{\pi_1(M)} V \to M$ and take the sheaf of sections. For (3) $\to$ (1) we use that a locally constant sheaf defines a vector bundle such that the transition functions are locally constant. This defines the flat local sections as the constant sections determining the connection completely. The resulting de Rham complex computes sheaf cohomology since it is a soft resolution by the Poincaré Lemma.

Now, we want to consider the higher categorical analogues of the classical situation. Roughly, this means replacing vector bundles and connections by dg vector bundles, the fundamental groupoid by the fundamental ∞-groupoid, sheaves of vector spaces by sheaves of chain complexes.

[3] We could also take R-modules for any ring R and get twists of R-cohomology.

(1) *Differential Geometry.* Let $E = \bigoplus_i E_i \to M$ be a $\mathbb{Z}$-graded vector bundle. The space $\Omega^*(M;E)$ has a 'diagonal' total $\mathbb{Z}$-grading by the sum of degrees. We define a $\mathbb{Z}$-graded connection to be a degree 1 operator $D : \Omega(M;E) \to \Omega(M;E)$. We will require *flatness*, i.e. $D^2 = 0$. The cohomology of the cochain complex $\Omega(M;E)$ is the twisted cohomology.

In parts of the literature this is also called a representation up to homotopy of the Lie algebroid TM.

(2) *Topology.* We define an ∞-local system to be a representation of the fundamental ∞-groupoid, i.e. a functor $L : \Pi_\infty(M) \to \mathcal{D}(R)$. [4] For a field $R = k$ the right hand side is just the ∞-category of chain complexes $\mathrm{Ch}(k)$.

(3) *Sheaf Theory.* This is again the most concise definition. A local system on M is a locally constant ∞-sheaf with values in some appropriate ∞-category $\mathcal{C}$. An appropriate choice is the derived ∞-category $\mathcal{D}(R)$ of a ring. By ∞-sheaf we mean a functor $\mathcal{F} : \mathcal{O}\mathrm{pen}(M)^{\mathrm{op}} \to \mathcal{C}$ satisfying a homotopy descent condition, i.e.

$$\mathcal{F}(U) \xrightarrow{\simeq} \mathrm{holim}(\textstyle\prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_{ij}) \mathrel{\substack{\rightarrow\\[-1ex]\rightarrow\\[-1ex]\rightarrow}} \prod_{i,j,k} \mathcal{F}(U_{ijk}) \mathrel{\substack{\rightarrow\\[-1ex]\rightarrow\\[-1ex]\rightarrow\\[-1ex]\rightarrow}} \dots)$$

The cohomology of this ∞-local system is the cohomology of the chain complex $\mathcal{F}(M)$.

(4) *Stable Homotopy Theory.* By the Brown representability theorem, any generalized cohomology theory E^* gives rise to an Ω-spectrum E, i.e. a sequence of spaces E_i with $E_i \simeq \Omega E_{i+1}$, such that the E-cohomology is given by homotopy classes of maps $E^i(M) = [M, E_i]$. We can also write this in terms of homotopy classes of maps of spectra as $[\Sigma^{\infty-i}_+ M, E] = \pi_i \mathcal{M}\mathrm{ap}_{Sp}(\Sigma^\infty_*(M), E)$. For ordinary cohomology with values in a ring R we denote this Eilenberg-MacLane ring spectrum by HR. A twist of E-cohomology is given by a functor

$$\Pi_\infty(M) \longrightarrow \mathcal{M}\mathrm{od}(E)\,.$$

Maybe the following: Note that $\mathcal{M}\mathrm{od}(E)$ is a delooping BE of E. (as is $Pic(E)$) Morally, it hence classifies E-bundles over M by taking the homotopy fiber F of the classifying map.

$$F \xrightarrow{\pi} \Pi_\infty(M) \longrightarrow \mathcal{M}\mathrm{od}(E)$$

The twisted cohomology is given by the homotopy groups of the space of sections. This is in turn the following homotopy pullback:

$$\begin{array}{ccc} \Gamma(F) & \longrightarrow & \mathcal{M}\mathrm{ap}(\Pi_\infty(M), F) \\ \downarrow & & \downarrow{\scriptstyle \pi_*} \\ * & \xrightarrow{\mathrm{id}} & \mathcal{M}\mathrm{ap}(\Pi_\infty(M), \Pi_\infty(M)) \end{array}$$

This diagram says that a section is a map that composes to the identity via π.

(5) *Geometric Field Theories.* Do I want this here?

[4] We should distinguish between the smooth ∞-groupoid and the singular ∞-groupoid. However, they are equivalent as ∞-categories.

Remark 3.0.1. *Why do we need to use (co)chain complexes as targets in* (2) *and* (3)*? Note that we cannot take the* 1*-category* $\mathcal{V}\mathrm{ec}$ *on the right hand side since the datum of such a functor is equivalent to a functor* $\Pi_1(M) = \mathrm{Ho}(\Pi_\infty(M)) \to \mathcal{V}\mathrm{ec}$*, i.e. an ordinary local system. The same thing happens in* (1)*, where we get ordinary connections by restricting to vector spaces. Item (4) can of course only be formulated appropriately in* ∞*-categories.*

Let us say a few words about the relation between these different viewpoints. Item (5) is formulated with the benefit of hindsight. The most general one is (4). Restricting to Eilenberg-MacLane ring spectra HR, we get $\mathcal{M}\mathrm{od}(HR) \simeq \mathcal{D}(R)$ relating this immediately to (2). The equivalence between (2) and (3) is not very surprising anymore. It is just the statement that locally constant sheaves with values in $\mathcal{D}(R)$ are the same as functors $\Pi_\infty(M) \to \mathcal{D}(R)$. There are many sources for this, e.g. **??**. The equivalence between (1) and (2) in the case of real coefficients is finally contained in the works of Block-Smith [] and Abad-Schätz [].

In the next section we discuss precise definitions of the twists that we are considering.

4. Defining the relevant dg-categories

This section serves to define the categories in Thm ? and ?.

4.1. **Differential Geometry.** Here we define the dg-categories of flat $\mathbb{Z}$-graded connections and flat $\mathbb{Z}/2$-graded connections on M. In the literature both are referred to as *superconnections*. In line with the terminology of supergeometry, a superconnection is a $\mathbb{Z}/2$-graded connection for us. Also, these categories are often referred to as the dg-categories of *representations up to homotopy*.

Definition 4.1.1. *Let* M *be a manifold. Let* $E = \bigoplus_{n\in\mathbb{Z}} E_n \to M$ *be a* $\mathbb{Z}$*-graded vector bundle over* M*. We require it to have totally finite rank and to be complex. The* E*-valued forms* $\Omega(M;E) = \bigoplus \Omega^m(M;E_n)$ *have a total grading given by* $m \pm n$*. A* $\mathbb{Z}$-graded connection *is an operator* D *on* $\Omega(M;E)$ *of total degree* 1*. It needs to satisfy the Leibniz rule*

$$D(\omega\sigma) = d\omega\sigma + (-1)^{|\omega|}\omega D\sigma\,.$$

It is flat if $D^2 = 0$*, i.e. if it defines a differential.*

Definition 4.1.2. *Let* M *be a supermanifold. Let* $E = E^{\underline{0}} \oplus E^{\underline{1}} \to M$ *be a super vector bundle. The* E*-valued forms* $\Omega(M;E)$ *have a* $\mathbb{Z}/2$*-grading given by total degree. A* superconnection *is an odd operator* $\nabla\!\!\!\!\nabla$ *on* $\Omega(M;E)$*. It needs to satisfy the Leibniz rule*

$$\nabla\!\!\!\!\nabla(\omega\sigma) = d\omega\sigma + (-1)^{|\omega|}\omega\nabla\!\!\!\!\nabla\sigma\,.$$

It is flat if $\nabla\!\!\!\!\nabla^2 = 0$*.*

Lemma 4.1.3. *Locally, these graded connections can be written as follows:*

1. *Let* E *be a trivial* $\mathbb{Z}$*-graded vector bundle. Any* $\mathbb{Z}$*-graded connection is given by* $D = d + A = d + A_0 + A_1 + \dots$ *where* $A_i \in \Omega(M;\mathrm{End}(E))^i$ *has total degree* i*. A is a total degree* 1 *endomorphism valued form.*
2. *Let* E *be a trivial super vector bundle. Any superconnection is given by* $\nabla\!\!\!\!\nabla = d + A$ *where* $A \in \Omega(M;\mathrm{End}(E))^{\underline{1}}$ *is an odd endomorphism valued form.*

We copy the definition from [AS18].

Definition 4.1.4. *The dg-category* $\mathsf{Flat}_{\mathbb{Z}}(M)$ *has as objects the flat* $\mathbb{Z}$*-graded vector bundles. The cochain complex of morphisms from* (E_1, D_1) *to* (E_2, D_2) *is given by* $\Omega(M; \mathrm{Hom}(E_1, E_2))$ *with its total grading. The differential of* φ *is given by*

$$d\varphi = D_2 \circ \varphi - (-1)^{|\varphi|}\varphi \circ D_1 \,. \tag{2}$$

Mimicking this, we get only a category enriched over $\mathrm{Ch}^{\mathbb{Z}/2}$ of super vector bundles. This is also done in [AS14].

Definition 4.1.5. *The* $\mathrm{Ch}^{\mathbb{Z}/2}$*-enriched category* $\mathsf{Flat}_{\mathbb{Z}/2}(M)$ *has objects the flat super vector bundles. The (periodic) cochain complex of morphisms* (E_1, ∇_1) *to* (E_2, ∇_2) *is given by* $\Omega(M; \mathrm{Hom}(E_1, E_2))$ *with its total* $\mathbb{Z}/2$*-grading. The differential of* φ *is given by* $d\varphi = \nabla_2 \circ \varphi - (-1)^{|\varphi|}\varphi\nabla_1$.

This can be regarded as a dg-category canonically where we hom spaces are 2-periodic. Technically, we are changing the enrichment along the lax monoidal functor $\mathrm{Ch}^{\mathbb{Z}/2} \to \mathrm{Ch}$. See section ?. Hence, we can talk about the dg-category $\mathsf{Flat}_{\mathbb{Z}/2}(M)$.

Question 4.1.6. *(1) What are the homotopy categories?*
(2) There is the collapse functor induced by $\mathbb{Z} \to \mathbb{Z}/2$.

4.2. $0|1$**-Field Theories.** Here we want to define the categories $0|1 - \mathrm{Twi}(M)$ and $0|1 - \mathrm{ETwi}(M)$ of twists of topological and euclidean $0|1$-Field theories. We also immediately relate them to their supergeometric incarnations. Let $B\mathrm{sVect}$ be a monoidal 2-category with the property that the category $\mathrm{Hom}(\mathbb{1}, \mathbb{1})$ of endomorphisms of its monoidal unit is isomorphic to sVect. There is a canonical choice for this delooping: the bicategory of algebras, bimodules and bimodule maps. In the $0|1$-case the precise choice will not matter, but in general there is a choice involved.

Definition 4.2.1. *A topological twist is a symmetric monoidal functor* $0|1{-}\mathsf{Bord}(M) \to B\mathrm{sVect}$. [5]

This definition is intentionally vague and we make it more precise now. We now sketch the reduction steps, as explicated in [HKST11]. Firstly, there is only one object in $0|1 - \mathsf{Bord}$, the monoidal unit and hence the image must lie entirely in $\mathrm{End}(\mathbb{1}) \simeq \mathrm{sVect}$. Secondly, the morphisms are generated by maps $\mathbb{R}^{0|1} \to M$ under disjoint union, all up to diffeomorphism. Hence, the twist is equivalent to a map $\mathrm{Hom}(\mathbb{R}^{0|1}, M) \to \mathrm{sVect}$. Include Diff. This suggests a better definition.

Definition 4.2.2. *A topological twist is a map of stacks* $\Pi TM // \mathrm{Diff}(\mathbb{R}^{0|1}) \to \mathrm{sVect}$. *A euclidean twist is a map of stacks* $\Pi TM // \mathrm{Iso}(\mathbb{R}^{0|1}) \to \mathrm{sVect}$.

Definition 4.2.3. *For any stack* $\mathfrak{X}$ *we call the groupoid* $\mathrm{Fun}_{\mathcal{S}\mathrm{Mfld}}(\mathfrak{X}, \mathrm{sVect})$ *the category of super vector bundles on* X. *Let* $\mathbb{1}$ *be the trivial rank one bundle and let* E *be a vector bundle on* $\mathfrak{X}$. *We call the vector space* $\Gamma(E) := \mathrm{Nat}_{\mathcal{S}\mathrm{Mfld}}(\mathbb{1}, E)$ *the space of sections of* E.

Remark 4.2.4. *We will model stacks as fibered categories and morphisms as fibered functors. To be very explicit, the fibered category* $\mathrm{sVect} \to \mathcal{S}\mathrm{Mfld}$ *has objects* $E \to M$ *and morphisms are pullback diagrams of vector bundles. The map to* $\mathcal{S}\mathrm{Mfld}$ *forgets the vector bundles. The fiber over* S *is the groupoid* $\mathrm{sVect}(S)$ *of vector bundles over* S.

[5] We did not yet include that everything should be fibered over $\mathcal{S}\mathrm{Mfld}$.

For any supergroup G acting on X the quotient stack $X//G$ is the stack whose fiber at S is the groupoid of principal G-bundles $P \to S$ together with a G-equivariant map $P \to X$. The morphisms need to intertwine these maps.

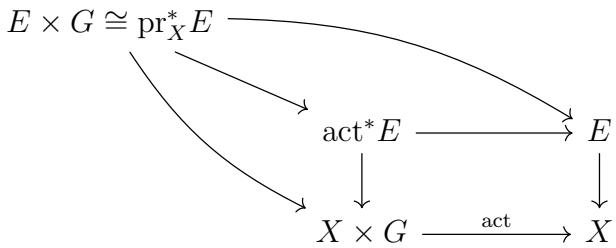
For quotient stacks we can do better. Of course all of the following remains true without the additional "super".

Definition 4.2.5. *Let X be a supermanifold with a G-action. A G-equivariant super vector bundle is a bundle $E \to X$ together with a fiberwise linear G-action compatible with the action on X.*

$$\begin{array}{ccc} E \times G & \xrightarrow{a_E} & E \\ \downarrow & & \downarrow \\ X \times G & \xrightarrow{a} & X \end{array}$$

They form a groupoid $\mathrm{sVect}^G(X)$ *whose morphisms are the equivariant isomorphisms.*

Let us do a little digression on equivalent definitions of what an equivariant vector bundle over a space X with G-action is. It is either defined as a fiberwise linear action on the bundle $E \times G \to E$ covering the action on the base or as a bundle map $a : \mathrm{pr}_X^* E \to \mathrm{act}^* E$ satisfying a cocycle equation. The equivalence is easy to see by the universal properties of pullbacks and depicted in the following diagram:



There is another sensible definition. Note that by the (2-categorical) Yoneda embedding we have $\mathrm{Fun}_{\mathrm{SMfld}}(S, \mathrm{sVect}) \simeq \mathrm{sVect}(S)$. Hence we can *define* a vector bundle over a stack $\mathfrak{X}$ to be a functor $\mathfrak{X} \to \mathrm{sVect}$. An equivariant vector bundle over a space X with a G-action is then a vector bundle over the quotient stack $X//G$. Note also that a section of a bundle $E \to S$ is the same as a bundle morphism from the trivial bundle $\mathbb{1} \to E$. We write $\mathbb{1}$ since it is the monoidal unit for the tensor product of bundles. Hence, for a bundle $E : \mathfrak{X} \to E$ we define $\Gamma(\mathfrak{X}, E) := \mathrm{Nat}_{\mathrm{SMfld}}(\mathbb{1}, E)$ to be the sections of the bundle.

Remark 4.2.6. *Since stacks are a $(2,1)$-category all natural transformations are invertible. This is of course unrealistic for a section. We need to allow also noninvertible maps of vector bundles, i.e. we regard E as a functor to the fibered category* sVect*. Formally, the inclusion of stacks into fibered categories has a left adjoint called stackification and a right adjoint $(-)^\circ$ that picks out the largest subcategory with only cartesian arrows. So the stack* sVect *is really the subcategory* sVect° *on cartesian arrows.*

Lemma 4.2.7. *There is an equivalence of categories* $\mathrm{Fun}_{\mathrm{SMfld}}(X//G, \mathrm{sVect}) \simeq \mathrm{sVect}^G(X)$. *It maps a functor $E : X//G \to$ sVect to its value on the principal bundle* $\mathrm{pr} : X \times G \to X$ *with the equivariant map to X given by the action.*

Furthermore, the sections of an equivariant bundle $E \to X$ are given by invariant sections.

Proof. Any principal bundle over S is locally trivial. The stack $X//G$ is the *stackification* of the prestack X/G whose fiber over $S \in \mathcal{S}\mathrm{Mfld}$ consists of only of *trivial* principal bundles $S \times G \to S$ with an equivariant map to X. Hence, $\mathrm{Fun}(X//G, \mathfrak{Y}) \simeq \mathrm{Fun}(X/G, \mathfrak{X})$ for any stack $\mathfrak{Y}$. The equivariant map $S \times G \to X$ is equivalently a map $f : S \to X$ exhibiting $S \times G$ as the pullback $f^*(X \times G \to X)$ of the trivial bundle over X whose map to X is the action.

$$\begin{array}{ccccc} S \times G & \longrightarrow & X \times G & \xrightarrow[\mathrm{act}]{} & X \\ \downarrow & & \downarrow & & \\ S & \xrightarrow{f} & X & & \end{array}$$

We now specify to $\mathfrak{Y} = \mathrm{sVect}$ and $\mathcal{E} : X/G \to \mathrm{sVect}$ a super vector bundle on $X//G$. Since fibered functors preserve pullbacks, the value of $\mathcal{E}$ is uniquely determined by the vector bundle $E := \mathcal{E}(P) \in \mathrm{sVect}(X)$, where $P = X \times G \to X$ is the trivial bundle. Additionally, we know that the pullbacks $\mathrm{pr}_X^* P$ and $\mathrm{act}^* P$ are isomorphic.

$$\begin{array}{ccccc} & & X & & \\ & \overset{(x,g,h)\mapsto x.gh}{\nearrow} & & \overset{(x,g,h)\mapsto x.h}{\nwarrow} & \\ \mathrm{pr}_X^* P \cong X \times G \times G & & \xrightarrow{\mathrm{id}\times\mathrm{mult}} & & X \times G \times G \cong \mathrm{act}^* P \\ & \searrow & & \swarrow & \\ & & X \times G & & \end{array}$$

This map $a : \mathrm{pr}_X^* P \to \mathrm{act}^* P$ satisfies the coherence that the following diagram of bundles over $X \times G \times G$ is commutative:

$$\begin{array}{ccccc} & & \mathrm{pr}^*\mathrm{pr}^* P & \xrightarrow{\mathrm{pr}^* a} & \mathrm{pr}^*\mathrm{act}^* P & & \\ & \swarrow\!\!\swarrow & & & \searrow\!\!\searrow & & \\ \mathrm{mult}^*\mathrm{pr}^* P & & & & & & \mathrm{act}^*\mathrm{pr}^* P \\ & \searrow^{\mathrm{mult}^* a} & & & & \swarrow_{\mathrm{act}^* a} & \\ & & \mathrm{mult}^*\mathrm{act}^* P & = & \mathrm{act}^*\mathrm{act}^* P & & \end{array}$$

Hence, the bundle E also comes with a map $a_E : \mathrm{pr}_X^* E \to \mathrm{act}^* E$ satisfying the coherence cocycle condition. This is precisely a G-equivariant vector bundle over X. Actually we spelled out explicitly what a vector bundle over a 2-coskeletal simplicial supermanifold is. In our case it is the (truncated) nerve of a (super) Lie groupoid.

$$X \times G \times G \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; X \times G \;\substack{\longrightarrow\\ \longrightarrow}\; X$$

Let us spell out what a fibered natural transformation $\sigma : \mathbb{1} \to \mathcal{E}$ is. For each S it assigns a map $\mathbb{1} \to \mathcal{E}(S)$, i.e. a section of the bundle $\mathcal{E}(S)$ that is natural in S. In particular, this is a section σ_P of $E = \mathcal{E}(P)$ and all the other sections are determined via pullback of this one. The section still needs to satisfy $a(\mathrm{pr}_X^* \sigma) = \mathrm{act}^* \sigma$ by naturality. This could be written as $g.\sigma(x) = \sigma(g.x)$. Hence σ is an invariant section. □

What about the orientation reversal? Probably not important. We now specialise to an action of the odd translation group $(\mathbb{R}^{0|1}, +)$ on X. The nature of the geometric odd line is very similar to an algebraic differential. We build a natural (2-periodic) dg-category whose homotopy category is $\mathrm{sVect}^{\mathbb{R}^{0|1}}(X)$. The following is an ad-hoc version. We will explicate a more systematic version below.

Definition 4.2.8. *The* $\mathrm{Ch}^{\mathbb{Z}/2}$*-enriched category* $\mathsf{sVect}^{\mathbb{R}^{0|1}}(X)$ *has objects the* $\mathbb{R}^{0|1}$*-equivariant vector bundles over* X*. The morphisms in even and odd degree are given by even and odd bundle homomorphisms that do* not *need to be equivariant.*

For $\varphi \in \mathrm{Hom}(E_1, E_2)$ *we define its differential* $d\varphi$ *as the derivative of a signed difference of the outer paths.*

$$\begin{array}{ccc} \mathbb{R}^{0|1} \times E_1 & \xrightarrow{\mathrm{id}\times\varphi} & \mathbb{R}^{0|1} \times E_2 \\ \downarrow{\scriptstyle a_1} & & \downarrow{\scriptstyle a_2} \\ E_1 & \xrightarrow{\varphi} & E_2 \end{array}$$

$$d\varphi = (-1)^{|\varphi|}\frac{\partial}{\partial\theta}\left(\varphi \circ a_1 - a_2 \circ (\mathrm{id} \times \varphi)\right) \tag{3}$$

The $\mathbb{Z}/2$-graded cochain complex of homomorphisms is the following picture.

$$\mathrm{Hom}(E_1, E_2)^{\underline{0}} \underset{d}{\overset{d}{\rightleftarrows}} \mathrm{Hom}(E_1, E_2)^{\underline{1}}$$

If we want to think of it as an ordinary $\mathbb{Z}$-graded cochain complex we can unfold it by setting $\mathrm{Hom}(E_1, E_2)^i = \mathrm{Hom}(E_1, E_2)^{\underline{i}}$:

$$\ldots \xrightarrow{d} \mathrm{Hom}(E_1, E_2)^{\underline{0}} \xrightarrow{d} \mathrm{Hom}(E_1, E_2)^{\underline{1}} \xrightarrow{d} \mathrm{Hom}(E_1, E_2)^{\underline{0}} \xrightarrow{d} \ldots$$

We now explicate the definition of the differential. $d\varphi$ should be thought of as the failure of φ to be equivariant.

Lemma 4.2.9. *d is a differential on* $\mathrm{Hom}(E_1, E_2)$*. We can write* $a_i = \mathrm{id} + \theta A_i$ *for odd bundle? maps* $A_i : E_i \to E_i$ *giving*

$$d\varphi = \varphi A_1 - (-1)^{|\varphi|} A_2 \varphi\,. \tag{4}$$

Proof. Note that the actions a_i can indeed be written as $\mathrm{id}+\theta A_i$ and the associativity of the action requires $A_i^2 = 0$:

$$(\mathrm{id} + \theta A_i)(\mathrm{id} + \eta A_i) = \mathrm{id} + (\theta + \eta)A_i - \theta\eta A_i^2\,.$$

We compute $\varphi \circ a_1 = \varphi + (-1)^{|\varphi|}\theta\varphi \circ A_1$ and $a_2 \circ (\mathrm{id} \times \varphi) = \varphi + \theta A_2 \circ \varphi$. Hence, the claimed formula for $d\varphi$ holds. d is indeed an odd differential:

$$\begin{aligned} d^2\varphi &= d\varphi A_1 - (-1)^{|\varphi|+1} A_2 d\varphi \\ &= \varphi A_1^2 - (-1)^{|\varphi|} A_2\varphi A_1 + (-1)^{|\varphi|} A_2\varphi A_1 + A_2^2\varphi = 0 \end{aligned}$$

□

To explain the systematic approach the following might be useful.

Example 4.2.10. *Let G be a group. We consider the category $G-\mathcal{S}\mathrm{et}$. Its objects are sets with a G-action. A morphism is a G-equivariant map. We can endow $G-\mathcal{S}\mathrm{et}$ with a self-enrichment. That is, we let* $\hom(X,Y)$ *be the set of* all *maps $X \to Y$ and the G-action is given by conjugation. The set of fixed points* $\hom(X,Y)^G$ *is precisely the set of morphisms in the ordinary category $G-\mathcal{S}\mathrm{et}$. This is the internal hom for the monoidal structure given by the product.*

Similarly, the category $\mathrm{sVect}^G(X)$ is enriched over itself. For two G-vector bundles E, E' this internal hom is given by the vector bundle of homomorphisms together with the conjugation action. We refer to the self-enriched category as $\mathsf{sVect}^G(X)$.

We apply this for $G = \mathbb{R}^{0|1}$. We saw that there is a functor $\Gamma : \mathrm{sVect}^{\mathbb{R}^{0|1}}(X) \to \mathrm{Ch}^{\mathbb{Z}/2}$ of global sections where the $\mathbb{R}^{0|1}$-action provides the differential. The global section functor is lax monoidal, see 4.5.3.

Hence, we have a canonical enrichment of $\mathsf{sVect}^{\mathbb{R}^{0|1}}(X)$ in $\mathrm{Ch}^{\mathbb{Z}/2}$ by taking global sections of the inner hom. The last step is two view $\mathrm{Ch}^{\mathbb{Z}/2} \subset \mathrm{Ch}$ as periodic cochain complexes. We abusively also denote this dg-category by $\mathsf{sVect}^{\mathbb{R}^{0|1}}(X)$.

4.3. **Equivariant vector bundles over ΠTM.** Here we discuss various variants of equivariant super vector bundles over ΠTM.

The $\mathbb{R}^{0|1}$-action on ΠTM is given by $a^*(\omega) = \omega + \theta d\omega$. [6] We write $\theta.\omega = \omega + \theta d\omega$ for better readability. [7] It is even part of an action by the monoid $\underline{\mathrm{Hom}}(\mathbb{R}^{0|1}, \mathbb{R}^{0|1}) \cong \mathbb{R} \rtimes \mathbb{R}^{0|1}$ given by $a^*(\omega) = t^{|\omega|}(\omega + \theta d\omega)$. Explicitly, we have the following multiplication:

$$1 \longrightarrow \mathbb{R}^{0|1} \longrightarrow \mathbb{R} \rtimes \mathbb{R}^{0|1} \longrightarrow \mathbb{R} \longrightarrow 1$$

$$(t_1, \theta_1)(t_2, \theta_2) = (t_1 t_2, \theta_1 + t_1 \theta_2)$$

The action of $\mathbb{R}$ on $\mathbb{R}^{0|1}$ is given by dilation $\rho_t(\theta) = t\theta$.

The subgroups that we are interested in are

$$\mathbb{R}^{0|1}, \qquad \{\pm 1\} \rtimes \mathbb{R}^{0|1}, \qquad \mathbb{R}^\times \rtimes \mathbb{R}^{0|1} \subset \mathbb{R} \rtimes \mathbb{R}^{0|1}$$

Let us fix a super vector bundle $E \to M$. Since the odd tangent bundle is homotopy equivalent to the base manifold M we can always assume that a bundle over ΠTM is given by $\mathcal{E} = \pi^* E$ for $\pi : \Pi TM \to M$ the projection. See 5.0.2 for the proof.

Case 1: $\mathbb{R}^{0|1}$. A $\mathbb{R}^{0|1}$-equivariant bundle map is a map $\mathrm{pr}^*\mathcal{E} \to \mathrm{act}^*\mathcal{E}$ satisfying associativity. Note that $\Gamma(\Pi TM, \mathcal{E}) = \Omega(M; E)$ Since we are identifying bundles with locally constant sheaves, this is equivalently a map $a_E^* : \Omega(M; E) \to \Omega(M; E)[\theta]$ such that for $\omega \in \Omega(M)$ and $\sigma \in \Omega(M; E)$ we have

$$a_E^*(\omega\sigma) = (a^*\omega)(a_E^*\sigma) = (\omega + \theta d\omega) a_E^* \sigma \,.$$

For $\theta = 0$ it must be the identity, so that $a_E^*\sigma = \sigma + \theta A(\sigma)$. A must be odd for the action to be even. From now on we write this abusively as $a_E(\sigma) = \sigma + \theta A(\sigma)$. The

[6] Maps between supermanifolds $S \to T$ may equivalently be given as pullbacks $C^\infty(T) \to C^\infty(S)$. In this case it is $\Omega(M) \to \Omega(M)[\theta]$.

[7] This can also be understood "literally" in the functor of points approach. That is, θ and ω are maps from a test space S to $\mathbb{R}^{0|1}$ and ΠTM respectively and we define a natural action $C^\infty(S, \mathbb{R}^{0|1}) \times C^\infty(S, \Pi TM) \to C^\infty(\Pi TM)$.

equivariance of the action is equivalent to the Leibniz rule for A since we write

$$a_E^*(\omega\sigma) = \omega\sigma + \theta A(\omega\sigma)\,,$$
$$(a^*\omega)(a_E^*\sigma) = (\omega + \theta d\omega)(\sigma + \theta A(\sigma)) = \omega\sigma + \theta(d\omega\sigma + (-1)^{|\omega|}\omega A(\sigma))\,.$$

This means A is a superconnection on E. Finally, flatness of A, i.e. $A^2 = 0$ is equivalent to associativity of the action.

$$\theta_1.\theta_2.\sigma = \theta_1.(\sigma + \theta_2 A(\sigma)) = \sigma + \theta_2 A(\sigma) + \theta_1 A(\sigma) + \theta_1\theta_2 A^2(\sigma)$$
$$(\theta_1 + \theta_2).\sigma = \sigma + (\theta_1 + \theta_2)A(\sigma)$$

Hence, the structure of an $\mathbb{R}^{0|1}$-equivariant vector bundle on $\mathcal{E} = \pi^*E$ over ΠTM is equivalent to a flat superconnection on E.

Case 2: $\{\pm 1\} \rtimes \mathbb{R}^{0|1}$. Still, the $\mathbb{R}^{0|1}$-action gives us a flat superconnection $\nabla\!\!\!\!\nabla$ on E. Denoting the generator of ± 1 by τ with $\tau^2 = 1$ we have that the conjugation action is given by $\tau\theta\tau^{-1} = -\theta$. The action of τ on $\mathcal{E}$ must also be compatible with the base action, i.e.

$$\tau.(\omega\sigma) = (-1)^{|\omega|}\omega(\tau.\sigma)\,.$$

In particular, τ acts C^∞-linearly on $\Gamma(E)$ and is hence given by an even bundle map $T \in \mathrm{End}(E)$ with $T^2 = \mathrm{id}$. [8] For general $\sigma \in \Omega(M;E)$ we get $\tau.\sigma = (-1)^{|\sigma|}T(\sigma)$. We must have

$$\tau.\nabla\!\!\!\!\nabla\sigma = -\nabla\!\!\!\!\nabla\tau.\sigma\,,$$

as $\tau.\theta.\sigma = (-\theta).\tau.\sigma$. E splits into the ± 1-eigenspaces of T as $E = E_1 \oplus E_{-1}$. Beware that both of them still have a different super grading. Hence, the E-valued forms also split into $\Omega(M;E_1) \oplus \Omega(M;E_{-1})$. The superconnection preserves these. To see this note that a homogeneous $\sigma \in \Omega(M;E)$ takes values in $E_{\pm 1}$ if and only $\tau.\sigma = \pm(-1)^{|\sigma|}\sigma$. Now we can check that the same is true for $\nabla\!\!\!\!\nabla\sigma$:

$$\tau.\nabla\!\!\!\!\nabla\sigma = -\nabla\!\!\!\!\nabla\tau.\sigma = \pm(-1)^{|\sigma|+1}\nabla\!\!\!\!\nabla\sigma = \pm(-1)^{|\nabla\!\!\!\!\nabla\sigma|}\nabla\!\!\!\!\nabla\sigma\,.$$

Hence we get that $(E, \nabla\!\!\!\!\nabla) = (E_1, \nabla\!\!\!\!\nabla_1) \oplus (E_{-1}, \nabla\!\!\!\!\nabla_{-1})$ and τ acts on $E_{\pm 1}$ by $\tau.\sigma = \pm(-1)^{|\sigma|}\sigma$. We think of this as encoding that τ decides if an equivariant section is even or odd.

Case 3: $\mathbb{R}$. We want to show that an action of $\mathbb{R}$ is equivalent to an $\mathbb{N}$-grading on a vector bundle. If the $\mathbb{R}$-action on a super vector bundle is *even*, it refines the $\mathbb{Z}/2$-grading. A general $\mathbb{R}$-action gives a $\mathbb{Z} \times \mathbb{Z}/2$-grading.

The infinitesimal vector field $\mathcal{E}\mathrm{ul}$ induced by the $\mathbb{R}$-action is called *Euler vector field*. It acts as $\mathcal{E}\mathrm{ul}_p = \frac{d}{dt}|_{t=1} t.p$.

Lemma 4.3.1. *Let $\mathbb{R}_{\geq 0} \times V \to V$ be a smooth linear action of the multiplicative monoid $(\mathbb{R}_{\geq 0}, \cdot)$ on a finite dimensional vector space. Then, the vector space splits as $V \cong \oplus_{n\in\mathbb{N}} V_n$ where the action on V_n is given by the homothety $t.v = t^n \cdot v$. The Euler vector field $\mathcal{E}\mathrm{ul}$ acts on V_n by $\mathcal{E}\mathrm{ul}(v) = nv$, i.e. V_n is the eigenspace of E.*

It is crucial that the action is smooth and extends to 0. Note that we regard the vector field $\mathcal{E}\mathrm{ul}$ as a linear map $V \to V$.

Proof. The strictly positive part $(\mathbb{R}_{>0}, \cdot)$ is isomorphic to $(\mathbb{R}, +)$ via $t \mapsto e^t$. Any smooth $(\mathbb{R}, +)$-action is given by $s \mapsto e^{sA}$ for some matrix A. W.l.o.g. we may

[8] There might or might not be a subtletly involving odd functions with a minus sign!

assume that A is in complex Jordan normal form. Further we may assume A is a Jordan block only, i.e.

$$A = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & 1 & \\ & & \ddots & \ddots \\ & & & \lambda \end{pmatrix} \qquad e^{sA} = e^{s\lambda} \begin{pmatrix} 1 & s & \frac{s^2}{2!} & \frac{s^3}{3!} & \dots \\ & 1 & s & \frac{s^2}{2!} & \dots \\ & & \ddots & \ddots & \vdots \\ & & & 1 & s \\ & & & & 1 \end{pmatrix}$$

We substitute back $s = \log(t)$. Note that this does not extend smoothly to 0 if the matrix is larger than 1×1 since the term $\log(t)$ appears in the matrix. Hence we are left with 1×1-Jordan blocks acting by $e^{s\lambda} = t^\lambda$. But this function extends smoothly to zero if and only if $\lambda \in \mathbb{N}$. □

Corollary 4.3.2. *Let $\mathbb{R}_{\geq 0} \times E \to E$ be a fiberwise action of the multiplicative monoid $\mathbb{R}_{\geq 0}$ on a vector bundle $E \to M$. Then, $E \cong \oplus_{n\in\mathbb{N}} E_n$ splits as a direct sum and the action on E_n is given by homothety $t.v = t^n \cdot v$. E_n is also the n-eigenspace of $\mathcal{E}\mathrm{ul}$.*

Note that we regard the vector field $\mathcal{E}\mathrm{ul}$ on the vector bundle E as a bundle map $E \to E$.

Proof. We bootstrap the fiberwise eigenspace decomposition of $\mathcal{E}\mathrm{ul}$. The eigenvalues depend continuously on the point and take values in $\mathbb{N}$ by the above Lemma. Hence, the equation $\mathcal{E}\mathrm{ul}(v) = nv$ has a locally constant rank of solutions. Now $\mathcal{E}\mathrm{ul} - n$ is a smooth bundle maps with constant rank. Hence, $E_n := \ker(\mathcal{E}\mathrm{ul} - n)$ is a smooth subbundle. By the Lemma it follows that $t.v = t^n v$ for $v \in E_n$. □

Remark 4.3.3. *The continuity of the eigenspaces is not trivial. They are the roots of the smoothly varying characteristic polynomial. Parametrizing roots smoothly is the subject matter of "choosing roots smoothly, II". See also Rellich's counterexample.*

Now we consider actions of the monoid $(\mathbb{R}, \cdot)$. Note that the action of $\mathbb{R}$ is fixed if we know its restriction to an $\mathbb{R}_{\geq 0}$-action and the action of -1. We always want to assume that the action is *even*, i.e. that -1 acts by the parity involution.

Of course we are interested in ΠTM. The action of $(\mathbb{R}, \cdot)$ is given by

$$t.\omega = t^{|\omega|}\omega\,.$$

Let $\mathcal{E} = \pi^* E$ be the pullback of a super vector bundle on M to ΠTM. Let $\mathcal{E}$ be equipped with an even $(\mathbb{R}, \cdot)$-action covering the above action on ΠTM. The zero section $z : M \to \Pi TM$ is equivariant where M has the trivial action. Hence, $E = z^*\mathcal{E}$ also carries an even $(\mathbb{R}, \cdot)$-action. This decomposes $E \cong \oplus_{n\in\mathbb{N}} E_n$ as a super vector bundle. [9] Let $v \in E_n$. Note that $(-t).v = -1.t.v = (-1)^{|v|}t^n v$, $t \geq 0$. At $t = 0$ left and right sided derivatives have to match, i.e. $(-1)^{|v|}n! = (-1)^n n!$ and hence the $\mathbb{N}$-grading refines the $\mathbb{Z}/2$-grading. In view of that we obtain $t.v = t^n v$ for all $t \in \mathbb{R}$.

Corollary 4.3.4. *Let $E \to M$ be a super vector bundle over a manifold and let $\mathcal{E} = \pi^* E \to \Pi TM$ equipped with an action of the monoid $(\mathbb{R}, \cdot)$ covering the trivial action on the base. Then $E \cong \oplus_{n\in\mathbb{N}} E_n$ and $t.\sigma = t^n\sigma$ on sections of E_n.*

[9] A super vector bundle on a purely even supermanifold is just a bundle with a parity involution. By definition, the action commutes with the parity involution. Hence, the decomposition is a decomposition as a super vector bundle.

Question 4.3.5. *What about supermanifolds M? what is the action on ΠTM, i.e. pseudo differential forms.*

Case 4: $\mathbb{R} \rtimes \mathbb{R}^{0|1}$. Recall that $E \to M$ is a super vector bundle with $\mathcal{E} = \pi^* E \to \Pi TM$ its pullback. Suppose that this comes equipped with an *even* action of the monoid $\mathbb{R} \rtimes \mathbb{R}^{0|1}$. The $\mathbb{R}^{0|1}$-action gives a flat superconnection $\nabla\!\!\!/$ on E. The $(\mathbb{R}, \cdot)$-action gives an $\mathbb{N}$-grading on E that we denote by deg. To unambiguously define the action on the semidirect product we need the following computations to be equal.

$$(t,\theta).\sigma = (1,\theta).(t,0).\sigma = t^{\deg \sigma}(\sigma + \theta \nabla\!\!\!/ \sigma)$$
$$(t,\theta).\sigma = (t,0).(1,\theta/t).\sigma = t^{\deg \sigma}\sigma + \theta t^{\deg(\nabla\!\!\!/\sigma)-1} \nabla\!\!\!/ \sigma$$

Hence, $\deg(\nabla\!\!\!/\sigma) = \deg \sigma + 1$ and the superconnection refines to a degree $+1$ operator on the complex $\Gamma(M, \mathcal{E}) = \Omega(M, E)$ with its total grading. That is, $\nabla\!\!\!/$ is a flat $\mathbb{N}$-graded connection.

Case ?: Recovering a $\mathbb{Z}$-grading Suppose we are given a $\mathbb{Z}$-graded bundle $E = \oplus_n E_n \to M$ with a flat $\mathbb{Z}$-graded connection D. Then, we can define a $\mathbb{R}^\times \rtimes \mathbb{R}^{0|1}$-equivariant vector bundle over ΠTM by taking $\mathcal{E} = \pi^* E$ and the action given on $\Gamma(\mathcal{E}) = \Omega(M; E)$ by letting

$$(t,\theta).\sigma = t^{\deg \sigma}(\sigma + \theta D\sigma)\,.$$

Note that this does not extend to zero. But there are also actions of $\mathbb{R}^\times \rtimes \mathbb{R}^{0|1}$ that are not of this form. The easiest example is to set $M = *$ with E a one-dimensional vector space. Note that $\Omega(M; E) \cong E$. An action is now just a smooth group homomorphism $\mathbb{R}^\times \to \mathbb{R}^\times$. These are generated by $t \mapsto |t|^\lambda$ and $t \mapsto \mathrm{sign}(t)$. In particular, there are not only integer exponents occuring.

What we really want to have is a group $\mathbb{G}_m$ such that actions of it correspond to a $\mathbb{Z}$-grading. The solution in algebraic geometry is to take $\mathbb{G}_m = \mathrm{Spec}(\mathbb{Z}[t, t^{-1}])$, the multiplicative group scheme. The set of maps from $\mathrm{Spec}(R)$ into $\mathbb{G}_m$ is the set of maps $\mathbb{Z}[t, t^{-1}] \to R$, i.e. the set of invertible elements in R. It acquires a group structure by multiplication. On the level of manifolds, the invertible functions $C^\infty(M)^\times$ are representable by (functions into) $\mathbb{R}^\times$. Then the inclusion $\mathbb{Z}[t, t^{-1}] \subset C^\infty(\mathbb{R}^\times)$ corresponds to a homomorphism $\mathbb{R}^\times \to \mathbb{G}_m$. Thus, a $\mathbb{Z}$-grading in particular gives a $\mathbb{R}^\times$-action. The converse is not true as we have seen.

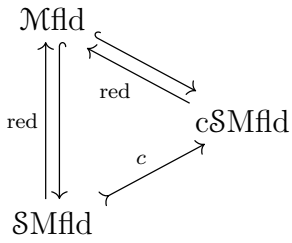
We need to go to *complex* functions and complex supermanifolds. Here, it is *not* true that $S \mapsto C^\infty(S)^\times$ is representable. The upshot is that this really yields the multiplicative group scheme. In particular it only has endomorphisms given by exponentiating by integers.

4.4. csManifolds and representability of the multiplicative group scheme.

Our setup now changes a bit. We need to complexify our local models. In particular, this means we are working in $\mathrm{sVect}_{\mathbb{C}}$. We should emphasize that we are *not* studying complex analytic supermanifold.

Definition 4.4.1. *Let $\mathbb{R}^{p|q}_{\mathrm{cs}} = (\mathbb{R}^p, \mathcal{O}_{\mathbb{R}^{p|q}_{\mathrm{cs}}})$ where $\mathcal{O}_{\mathbb{R}^{p|q}}(U) = C^\infty(U, \mathbb{C}) \otimes \bigwedge^*_{\mathbb{C}}(\theta_1, \ldots, \theta_q)$. A complex supermanifold (cs manifold) is a pair $(|S|, \mathcal{O}_S)$ that is locally isomorphic to $\mathbb{R}^{p|q}_{\mathrm{cs}}$. $|S|$ needs to be a topological manifold. The morphisms are maps of locally ringed spaces. Denote the category of cs manifolds by* cSMfld.

Note that there is a faithful [10] complexification functor c and reduction functors that fit into the following diagram:



Reduction is right adjoint to the inclusion. Since $M \cong c(M)_{\mathrm{red}}$ for $M \in \mathcal{M}\mathrm{fld}$ we will regard $\mathcal{M}\mathrm{fld} \subset \mathrm{c}\mathcal{S}\mathrm{Mfld}$ as a full subcategory. The complexification is given by complexifying the structure sheaf and maps $c(\mathbb{R}^{p|q}) = \mathbb{R}^{p|q}_{\mathrm{cs}}$. There is also a functor Π from *complex* vector bundles over ordinary manifolds to $\mathrm{c}\mathcal{S}\mathrm{Mfld}$ defined as in the real case. Importantly, Batchelor's theorem holds true, i.e. for every complex supermanifold there is a complex vector bundle E such that it is isomorphic to ΠE. An important example is the bundle $\Pi T_{\mathbb{C}} M$ for $M \in \mathcal{M}\mathrm{fld}$ whose sheaf of functions is $\Omega_{\mathbb{C}}(M)$. This also agrees with $c(\Pi TM)$. Another result that carries over is that $\mathrm{c}\mathcal{S}\mathrm{Mfld}(S, T) \cong \mathrm{s}\mathcal{A}\mathrm{lg}(\mathcal{O}(T), \mathcal{O}(S))$.

Passing to complex supermanifolds we lose an important property. In the category $\mathcal{S}\mathrm{Mfld}$ it is easy to construct morphisms to $\mathbb{R}^{p|q}$ since $\mathbb{R}$ represents $S \mapsto \mathcal{O}(S)^{\mathrm{even}}$ and $\mathbb{R}^{0|1}$ represents $S \mapsto \mathcal{O}(S)^{\mathrm{odd}}$. The analogous result for cs manifolds is slightly more complicated, c.f. 4.4.6. We first state a partially negative result.

Lemma 4.4.2. *(1) The functor $S \mapsto \mathcal{O}(S)^{\mathrm{odd}}$ is represented by $\mathbb{R}^{0|1}_{\mathrm{cs}}$.*
(2) For $M \in \mathcal{M}\mathrm{fld}$ the internal hom $\underline{\mathrm{c}\mathcal{S}\mathrm{Mfld}}(\mathbb{R}^{0|1}_{\mathrm{cs}}, M)$ is represented by $\Pi T_{\mathbb{C}} M$.
(3) The functor $S \mapsto \mathcal{O}(S)^{\mathrm{even}}$ for $S \in \mathrm{c}\mathcal{S}\mathrm{Mfld}$ is not representable.
(4) The internal hom $\underline{\mathrm{c}\mathcal{S}\mathrm{Mfld}}(\mathbb{R}^{0|1}_{\mathrm{cs}}, \mathbb{R}^{0|1}_{\mathrm{cs}})$ is not representable. prove this maybe.

Proof. For the first assertion just note that $\mathcal{O}(S)^{\mathrm{odd}}$ is in bijection with even homomorphisms $\mathbb{C}[\theta]/(\theta^2) \to \mathcal{O}(S)$. The proof of the second assertion is the same as in the real case, c.f. []. For the third assertion note that $\mathrm{c}\mathcal{S}\mathrm{Mfld}(S \times \mathbb{R}^{0|1}_{\mathrm{cs}}, \mathbb{R}^{0|1}_{\mathrm{cs}}) \cong \mathcal{O}(S \times \mathbb{R}^{0|1}_{\mathrm{cs}})^{\mathrm{odd}}$.

Suppose $S \mapsto \mathcal{O}(S)^{\mathrm{even}}$ was representable by $X \in \mathrm{c}\mathcal{S}\mathrm{Mfld}$. Then, restricted to $M \in \mathcal{M}\mathrm{fld}$ we get

$$\mathcal{M}\mathrm{fld}(M, \mathbb{C}) \cong \mathcal{O}(M)^{\mathrm{even}} \cong \mathrm{c}\mathcal{S}\mathrm{Mfld}(M, X) \cong \mathcal{M}\mathrm{fld}(M, X_{\mathrm{red}})\,.$$

Hence, $X_{\mathrm{red}} \cong \mathbb{C}$ and by Batchelor's theorem $X \cong \Pi E$ for $E \to \mathbb{C}$ a complex vector bundle. E must be trivial and thus $X \cong \mathbb{C} \times \mathbb{R}^{0|q}_{\mathrm{cs}}$. But the second factor represents q-tuples of odd functions, i.e. we must have $q = 0$. To conclude we argue by contradiction. The internal hom $\underline{\mathrm{c}\mathcal{S}\mathrm{Mfld}}(\mathbb{R}^{0|1}_{\mathrm{cs}}, \mathbb{R}^{0|1}_{\mathrm{cs}})$ would be represented by $\mathbb{C} \times \mathbb{R}^{0|1}_{\mathrm{cs}}$. To see this we proved that $\mathrm{c}\mathcal{S}\mathrm{Mfld}(S \times \mathbb{R}^{0|1}_{\mathrm{cs}}, \mathbb{R}^{0|1}_{\mathrm{cs}})$ is naturally isomorphic to $\mathcal{O}(S \times \mathbb{R}^{0|1}_{\mathrm{cs}})^{\mathrm{odd}} \cong \mathcal{O}(S)^{\mathrm{odd}} \oplus \theta\mathcal{O}(S)^{\mathrm{even}}$. By assumption this is represented by $\mathbb{R}^{0|1}_{\mathrm{cs}} \times X \cong \mathbb{R}^{0|1}_{\mathrm{cs}} \times \mathbb{C}$. Now consider $M \times \mathbb{R}^{0|2}_{\mathrm{cs}}$ for $M \in \mathcal{M}\mathrm{fld}$. Its even functions are

[10] It is not full: E.g. for $\mathbb{R}^{0|1} \in \mathcal{S}\mathrm{Mfld}$ there is the morphism $\mathbb{R}^{0|1}_{\mathrm{cs}} \to \mathbb{R}^{0|1}_{\mathrm{cs}}$ mapping $\theta \mapsto i\theta$ which is not the complexification of a morphism of supermanifolds. The image are the *real* morphisms, i.e. those that are invariant under complex conjugation.

given by $C^\infty(M,\mathbb{C}) \oplus \theta_1\theta_2 C^\infty(M,\mathbb{C})$. This means that the restriction of the funtor $\mathcal{O}(M \times \mathbb{R}^{0|2}_{\mathrm{cs}})^{\mathrm{even}}$ to $M \in \mathcal{M}\mathrm{fld}$ is represented by $\mathbb{C}^2$. However, we can compute

$$\begin{aligned}
\mathrm{c}\mathcal{S}\mathrm{Mfld}(M \times \mathbb{R}^{0|2}_{\mathrm{cs}}, \mathbb{C}) &\cong \mathrm{c}\mathcal{S}\mathrm{Mfld}(M, \underline{\mathrm{c}\mathcal{S}\mathrm{Mfld}}(\mathbb{R}^{0|1}_{\mathrm{cs}}, \Pi T_{\mathbb{C}}\mathbb{C}) \\
&\cong \mathrm{c}\mathcal{S}\mathrm{Mfld}(M, \underline{\mathrm{c}\mathcal{S}\mathrm{Mfld}}(\mathbb{R}^{0|1}_{\mathrm{cs}}, \mathbb{C}) \times \underline{\mathrm{c}\mathcal{S}\mathrm{Mfld}}(\mathbb{R}^{0|1}_{\mathrm{cs}}, \mathbb{R}^{0|2}_{\mathrm{cs}})) \\
&\cong \mathrm{c}\mathcal{S}\mathrm{Mfld}(M, \mathbb{C} \times \mathbb{R}^{0|2}_{\mathrm{cs}} \times (\mathbb{C} \times \mathbb{R}^{0|1}_{\mathrm{cs}})^2) \\
&\cong \mathcal{M}\mathrm{fld}(M, \mathbb{C}^3)
\end{aligned}$$

We arrive at the contradiction $\mathbb{C}^2 \cong \mathbb{C}^3$. □

Definition 4.4.3. *We call the functor*

$$\begin{aligned}
D : \mathrm{c}\mathcal{S}\mathrm{Mfld}^{\mathrm{op}} &\longrightarrow \mathsf{Grp} \\
S &\longmapsto (\mathcal{O}(S)^{\mathrm{even}})^\times
\end{aligned}$$

the complex dilation group or multiplicative group scheme.

The same proof as above also shows that D is not representable, whereas $D \circ c$ is representable by the multiplicative Lie group $\mathbb{C}^\times$: It maps $S \in \mathcal{S}\mathrm{Mfld}$ to $(\mathcal{O}(S)^{\mathrm{even}} \otimes \mathbb{C})^\times \cong \mathcal{S}\mathrm{Mfld}(S, \mathbb{C}^\times)$.

Lemma 4.4.4 ([Bor10])**.** *The endomorphisms of D are given by*

$$\begin{aligned}
\mathcal{O}(S)^{\mathrm{even}} &\longrightarrow \mathcal{O}(S)^{\mathrm{even}} \\
f &\longmapsto f^n
\end{aligned}$$

for $n \in \mathbb{Z}$.

Proof. Firstly, D composed with c on $\mathcal{S}\mathrm{Mfld}$ is represented by $\mathbb{C}^\times \cong S^1 \times \mathbb{R}$. The monoid of endomorphisms of this group is the multiplicative monoid $\mathbb{Z} \times \mathbb{C} \ni (n, c)$ acting by $re^{i\varphi} \mapsto r^c e^{in\varphi}$. We can also write this as $z \mapsto z^{\frac{c+n}{2}} \overline{z}^{\frac{c-n}{2}}$. Now any endomorphism $T : D \to D$ restricted to $\mathcal{S}\mathrm{Mfld}$ is given by such a pair (n, c). But any $S \in \mathrm{c}\mathcal{S}\mathrm{Mfld}$ is locally isomorphic to $c(\mathbb{R}^{p|q})$. This already determines T uniquely by naturality. The last thing to show is that necessarily $c = n$. For this we consider $\psi : \mathbb{R}^{0|2}_{\mathrm{cs}} \to \mathbb{R}^{0|2}_{\mathrm{cs}}$ given by $\psi^* : \theta_1 \mapsto \theta_1, \theta_2 \mapsto i\theta_2$. This is a morphism that is not in the image of c. We consider the function $f = 1 + \theta_1\theta_2 \in (\mathcal{O}(\mathbb{R}^{0|2}_{\mathrm{cs}})^{\mathrm{even}})^\times$. Naturality of T demands that $(\psi^* f)^{\frac{c+n}{2}} \overline{\psi^* f}^{\frac{c-n}{2}} = \psi^*(f^{\frac{c+n}{2}} \overline{f}^{\frac{c-n}{2}})$.

$$\begin{aligned}
(\psi^* f)^{\frac{c+n}{2}} \overline{\psi^* f}^{\frac{c-n}{2}} &= (1 + i\theta_1\theta_2)^{\frac{c+n}{2}} (1 - i\theta_1\theta_2)^{\frac{c-n}{2}} \\
&= \left(1 + i\frac{c+n}{2}\theta_1\theta_2\right)\left(1 - i\frac{c-n}{2}\theta_1\theta_2\right) = (1 + in\theta_1\theta_2) \\
\psi^*\left(f^{\frac{c+n}{2}} \overline{f}^{\frac{c-n}{2}}\right) &= \psi^*\left(\left(1 + \frac{c+n}{2}\theta_1\theta_2\right)\left(1 + \frac{c-n}{2}c\theta_1\theta_2\right)\right) \\
&= \psi^*(1 + c\theta_1\theta_2) = (1 + ic\theta_1\theta_2)
\end{aligned}$$

Hence $c = n$. Every endomorphism of D must be given by $f \mapsto f^n$. Clearly, this is indeed a multiplicative endomorphism. □

This is basically what we wanted. The goal is now to prove:

Theorem 4.4.5. *Let $X \in \mathcal{M}\mathrm{fld}$ be equipped with the trivial D-action. Then a D-equivariant action on a bundle $E \to X$ is equivalent to a $\mathbb{Z}$-grading.*

The following is a result that carries over from $\mathfrak{S}\mathrm{Mfld}$ with a slight modification.

Lemma 4.4.6. *Let $U \subset \mathbb{R}^{p|q}_{\mathrm{cs}}$ be open and $S \in \mathfrak{S}\mathrm{Mfld}$. Then, there is a canonical bijection between $\varphi \in \mathrm{c}\mathfrak{S}\mathrm{Mfld}(S, U)$ and*

$$\{(f_1, , \ldots, f_p, g_1, \ldots, g_q) : f_i \in \mathcal{O}(S)^{\mathrm{even}}, g_j \in \mathcal{O}(S)^{\mathrm{odd}}, (f_i)_{\mathrm{red}}(S_{\mathrm{red}}) \subset U_{\mathrm{red}}\}$$

given by $f_i = \varphi^*(x_i)$ and $g_j = \varphi^*(\theta_j)$. In particular, the $(f_i)_{\mathrm{red}}$ need to be real-valued.

Example 4.4.7. *Consider $\varphi : \mathbb{R}^{0|2}_{\mathrm{cs}} \to \mathbb{C}^\times$. We can write $\varphi^* x = x_0 + \lambda_1\theta_1\theta_2$ and $\varphi^* y = y_0 + \lambda_1\theta_2\theta_2$. We must have $z_0 = x_0 + iy_0 \in \mathbb{C}^\times$. Hence we can identify $\mathrm{c}\mathfrak{S}\mathrm{Mfld}(\mathbb{R}^{0|2}_{\mathrm{cs}}, \mathbb{C}^\times)$ with $\mathbb{C}^\times \times \mathbb{C}^2$ via $\varphi \mapsto (z_0, \lambda_1, \lambda_2)$. We cannot write φ as $z_0 + \lambda\theta_1\theta_2$ which is related to the above results on non-representability. What we can do is to consider $\varphi^* z := \varphi^* x + i\varphi^* y$. The mapping $\varphi \mapsto \varphi^* z \in (\bigwedge_{\mathbb{C}}(\theta_1, \theta_2)^\times)^{\mathrm{even}}$ is a group homomorphism. Identifying the latter with $\mathbb{C}^\times \times \mathbb{C}$ via $(z, \lambda) \mapsto z + \lambda\theta_1\theta_2$ we get the map*

$$\begin{aligned} \mathrm{c}\mathfrak{S}\mathrm{Mfld}(\mathbb{R}^{0|2}_{\mathrm{cs}}, \mathbb{C}^\times) \cong \mathbb{C}^\times \times \mathbb{C}^2 &\longrightarrow \mathbb{C}^\times \times \mathbb{C} \cong \left(\mathcal{O}(\mathbb{R}^{0|2}_{\mathrm{cs}})\right)^\times \\ (z_0, \lambda_1, \lambda_2) &\longmapsto (z_0, \lambda_1 + i\lambda_2)\,. \end{aligned}$$

Note that this is an epimorphism. Let us add to this example the effect of changing coordinates on $\mathbb{C}^\times \cong \mathbb{R} \times S^1$ via $(x, y) = e^{t+i\varphi}$. The inverse is $(t, \varphi) = (\frac{1}{2}\log(x^2 + y^2), \arctan(\frac{y}{x}))$. Any map $\mathbb{R}^{0|2}_{\mathrm{cs}} \to \mathbb{C}^\times$ can still be expressed in terms of the pullbacks of t and φ, i.e. by the same arguments we get a different isomorphism $\mathrm{c}\mathfrak{S}\mathrm{Mfld}(\mathbb{R}^{0|2}_{\mathrm{cs}}, \mathbb{C}^\times) \cong \mathbb{C}^\times \times \mathbb{C}^2$. The effect that this transformation has on λ_1, λ_2 is calculated via a first order Taylor series expansion yielding

$$\begin{aligned} \mathbb{C}^\times \times \mathbb{C}^2 &\longrightarrow \mathbb{C}^\times \times \mathbb{C}^2 \\ ((x, y), \lambda_1, \lambda_2) &\longmapsto ((t, \varphi), (\lambda_1', \lambda_2')) \\ \begin{pmatrix} \lambda_1' \\ \lambda_2' \end{pmatrix} &= e^{-t} \begin{pmatrix} \cos(\varphi) & \sin(\varphi) \\ -\sin(\varphi) & \cos(\varphi) \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix} \end{aligned}$$

Particular elements in the kernel of the map to $D(\mathbb{R}^{0|2}_{\mathrm{cs}})$ are $(z_0, \lambda_1, \lambda_2) = (1, i, -1)$ corresponding to $t = 0, \varphi = 0$ and also $(\lambda_1', \lambda_2') = (i, -1)$.

There is a canonical group homomorphism $\mathbb{C}^\times \to D$. It is induced by $z \in D(\mathbb{C}^\times) = (\mathcal{O}(\mathbb{C}^\times)^{\mathrm{even}})^\times$, i.e. its functor of points is

$$\begin{aligned} \mathrm{c}\mathfrak{S}\mathrm{Mfld}(S, \mathbb{C}^\times) &\longrightarrow D(S) \\ f &\longmapsto f^*(z)\,. \end{aligned}$$

We have seen that $D \circ c : \mathfrak{S}\mathrm{Mfld} \to \mathsf{Grp}$ is represented by $\mathbb{C}^\times$. Hence, for $S \in \mathfrak{S}\mathrm{Mfld}$ we get the following diagram:

$$\begin{array}{ccc} \mathfrak{S}\mathrm{Mfld}(S, \mathbb{C}^\times) & \xrightarrow{\cong} & D(c(S)) \\ \downarrow & \nearrow & \\ \mathrm{c}\mathfrak{S}\mathrm{Mfld}(c(S), \mathbb{C}^\times) & & \end{array}$$

Since all cs manifolds are locally isomorphic to $c(S)$, the map $\mathbb{C}^\times \to D$ is an epimorphism of sheaves. Hence D is a quotient of $\mathbb{C}^\times$. That is, it is a property of a group homomorphism out of $\mathbb{C}^\times$ to factor through D. We now apply this to representations, i.e. maps $D \to \mathrm{GL}_n(\mathbb{C})$. By the Yoneda Lemma and since $\mathcal{M}\mathrm{fld} \to \mathrm{c}\mathfrak{S}\mathrm{Mfld}$

is fully faithful, the map $\mathbb{C}^\times \to D \to \mathrm{GL}_n(\mathbb{C})$ is a Lie group homomorphism. Any representation of $\mathbb{C}^\times$ is given by mapping $\rho : z = e^{t+i\varphi} \mapsto e^{tA+i\varphi B}$. Up to conjugation we may assume that $A \in \mathrm{GL}_n(\mathbb{C})$ is in Jordan normal form and B is a diagonal matrix with integer eigenvalues. A and B commute, hence B must be an integer multiple of the identity. Fix a representation of D, let $\rho : \mathbb{C}^\times \to \mathrm{GL}_n(\mathbb{C})$ be the corresponding representation of $\mathbb{C}^\times$ and w.l.o.g. we suppose that

$$A = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & 1 & \\ & & \ddots & \ddots \\ & & & \lambda \end{pmatrix}, \qquad B = \begin{pmatrix} k & & & \\ & k & & \\ & & \ddots & \\ & & & k \end{pmatrix}.$$

The following diagram must commute for $S \in \mathcal{S}\mathrm{Mfld}$, thereby uniquely determining the representation of D:

$$\begin{array}{ccc} \mathrm{c}\mathcal{S}\mathrm{Mfld}(c(S), \mathbb{C}^\times) & \twoheadrightarrow & D(c(S)) \\ & \searrow^{\rho_*} & \downarrow \\ & & \mathrm{c}\mathcal{S}\mathrm{Mfld}(c(S), \mathrm{GL}_n(\mathbb{C})) \end{array}$$

In example 4.4.7 we computed that for $S = \mathbb{R}^{0|2}$ the map $f : \mathbb{R}^{0|2}_{\mathrm{cs}} \to \mathbb{C}^\times$ corresponding to $z = 1$ and $\lambda_1' = i$, $\lambda_2' = -1$ lies in the kernel of the map to D. Hence, $\rho_*(f) = 0$. We can formally write the map f as $t = i\theta_1\theta_2$ and $\varphi = -\theta_1\theta_2$. Again, via Taylor expansion, the composite $\rho \circ f$ is seen to be

$$e^{i\theta_1\theta_2 A + i(-1)\theta_1\theta_2 B} = \mathrm{id} + \theta_1\theta_2(A - B)\,.$$

From that we get that $A = B = k\mathbb{1}$. The action must be diagonal with integer eigenvalues. All in all, $\rho(z) = \rho(t, \varphi) = \exp((t + i\varphi)k1) = z^k\mathbb{1}$. We have proved the following theorem.

Theorem 4.4.8. *The generalized cs Lie group D is a quotient of $\mathbb{C}^\times$. Any smooth representation of D on a finite dimensional complex vector space V is given by a grading $V = \bigoplus_{n\in\mathbb{Z}} V_n$ and the corresponding action*

$$\begin{aligned} D(S) \times \mathrm{c}\mathcal{S}\mathrm{Mfld}(S, V_n) &\longrightarrow \mathrm{c}\mathcal{S}\mathrm{Mfld}(S, V_n) \\ (f, v) &\longmapsto f^n v\,. \end{aligned}$$

This action on V_n is induced by the $\mathbb{C}^\times$-action $z.v = z^n v$. The spaces V_n are characterized as the n-eigenspace of the Euler vector field $\mathcal{E}\mathrm{ul} = \frac{d}{dt}\big|_{t=0} e^t.v$.

We call an action of D *even* if -1 acts by parity involution.

Corollary 4.4.9. *Let $M \in \mathcal{M}\mathrm{fld}$ and $E \to M$ a complex super vector bundle with an even action by D covering the trivial action on M. Then, E acquires a $\mathbb{Z}$-grading refining the $\mathbb{Z}/2$-grading.*

Proof. We are dealing with a particularly nice $\mathbb{C}^\times$-action. Proceed as in the $\mathbb{R}^\times$-case. □

4.5. **Equivariant vector bundles over $\Pi T_{\mathbb{C}}M$.** Here, we discuss the construction of a dg-category of $\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})$-equivariant vector bundles over $\Pi T_{\mathbb{C}}M$ for $M \in \mathcal{M}\mathrm{fld}$.

Recall that the invertible part of $\underline{\mathrm{c}\mathcal{S}\mathrm{Mfld}}(\mathbb{R}^{0|1}_{\mathrm{cs}}, \mathbb{R}^{0|1}_{\mathrm{cs}})$ is isomorphic to $D \rtimes \mathbb{R}^{0|1}_{\mathrm{cs}}$ and we refer to the latter generalized super Lie group as $\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})$. Also, we discussed

that $\Pi T_{\mathbb{C}}M$ represents $\underline{\mathrm{c\mathcal{S}Mfld}}(\mathbb{R}^{0|1}_{\mathrm{cs}}, M)$. The induced action of $\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})$ on $\Pi T_{\mathbb{C}}M$ is given by the de Rham differential and the $\mathbb{Z}$-grading of $\Omega_{\mathbb{C}}(M)$. We could write

$$(z, \theta).\omega = z^{|\omega|}(\omega + \theta d\omega)\,.$$

It is slightly difficult to make sense of formulas like these. To clarify once, this last equation could literally be understood to hold in $\Omega_{\mathbb{C}}(M)$ for any fixed maps $z \in D(S)$ and $\theta : S \to \mathbb{R}^{0|1}$.

Let $E \to \Pi T_{\mathbb{C}}M$ be a $\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})$-equivariant complex super vector bundle. To clarify, we require such actions to be even. Then, the D-action equips $E = \oplus_n E_n$ with a $\mathbb{Z}$-grading corresponding to the eigenspaces of the infinitesimal generator $\mathcal{E}\mathrm{ul}$. Furthermore, the action on E_n is given by $z.\sigma = z^n\sigma$. The $\mathbb{R}^{0|1}_{\mathrm{cs}}$-action corresponds to a superconnection $\mathbb{\nabla}$. The superconnection becomes a degree $+1$ operator in the $\mathbb{Z}$-grading since

$$[\mathcal{E}\mathrm{ul}, \mathbb{\nabla}] = \mathbb{\nabla}\,.$$

Note that the category $\mathrm{sVect}_{\mathbb{C}}^{\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})}(X)$ is enriched over itself. That is, the vector bundle $\underline{\mathrm{Hom}}(E, E')$ carries a $\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})$-action by conjugation. Taking global sections of $\underline{\mathrm{Hom}}(E, E')$ we get cochain complex over $\mathbb{C}$.

Definition 4.5.1. *The dg-category* $\mathsf{sVect}_{\mathbb{C}}^{\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})}(X)$ *has objects the* $\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})$*-equivariant complex super vector bundles over* X*. The morphisms are the cochain complex described above.*

Explicitly, the morphisms $\mathrm{Hom}(E, E')_n$ *of degree* n *are those vector bundle homomorphisms* $\varphi : E \to E'$ *for which* $z.\varphi(\sigma) = z^n\varphi(z.\sigma)$*. The differential* d *is given by* $d\varphi(\sigma) = \frac{\partial}{\partial\theta}\theta.\varphi(-\theta.\sigma)$*.*

Lemma 4.5.2. *This indeed defines a dg-category.*

This in turn follows from the following because we just change enrichment to chain complexes.

Lemma 4.5.3. *The global section functor* $\Gamma : \mathrm{sVect}_{\mathbb{C}}^{\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})}(X) \to \mathrm{Ch}(\mathbb{C})$ *is lax monoidal.*

Proof. Aside from checking coherences, we need to provide a comparison map $\Gamma(E)_n \otimes \Gamma(E')_m \to \Gamma(E \otimes E')_{m+n}$ which is given by tensoring sections. Of course we have $z.(\sigma \otimes \sigma') = (z.\sigma) \otimes (z.\sigma') = z^{n+m}\sigma \otimes \sigma'$. This is a map of cochain complexes: The differential on the left and right hand side are respectively:

$$\begin{aligned} d(\sigma \otimes \sigma') &= \frac{\partial}{\partial\theta}\theta.\sigma \otimes \sigma' + (-1)^n\sigma \otimes \frac{\partial}{\partial\theta}\theta.\sigma' \\ d(\sigma \otimes \sigma') &= \frac{\partial}{\partial\theta}\left(\theta.\sigma \otimes \theta.\sigma'\right) \end{aligned}$$

They agree by the Leibniz rule. The monoidal unit in $\mathrm{sVect}_{\mathbb{C}}^{\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})}(X)$ is the trivial bundle $\underline{\mathbb{C}}$ with the trivial action. The comparison map between the monoidal units is the inclusion as constant sections $\varepsilon : \mathbb{C} \to \Gamma(\underline{\mathbb{C}})$. All coherences hold trivially. □

Lemma 4.5.4. *Let $E \to X$ be a $\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})$-equivariant vector bundle. Then $\Gamma(E)$ is a cochain complex with*

$$\Gamma(E)^n = \{\sigma \in \Gamma(E) : z.\sigma = z^n\sigma\}$$
$$d\sigma = \frac{\partial}{\partial\theta}\theta.\sigma$$

Proof. The differential is just the infinitesimal generator of the $\mathbb{R}^{0|1}_{\mathrm{cs}}$-action. It squares to zero. It maps $\Gamma(E)^n$ to $\Gamma(E)^{n+1}$. To check this we just differentiate $z.\theta.\sigma = z\theta.z.\sigma = z^{n+1}\theta.\sigma$. This gives $z.\frac{\partial}{\partial\theta}(\theta.\sigma) = z^{n+1}\frac{\partial}{\partial\theta}\theta.\sigma$. □

Remark 4.5.5. *We do not necessarily have that the vector space of all sections of E equipped with the structure of a cochain complex, i.e. not every section must be the sum of such homogeneous ones. Only in special cases $\Gamma(E)_n = \Gamma(E_n)$; i.e. for a trivial D-action on X. This is not true for $\Pi T_{\mathbb{C}}M$. There the form degree contributes, i.e. $\Omega(M;E)^n = \bigoplus_{k+l=n}\Omega^k(M)\otimes\Gamma(E_l)$*

5. Proofs

Proposition 5.0.1. *The categories* $\mathsf{Flat}_{\mathbb{Z}/2}(M)$ *and* $\mathsf{sVect}^{\mathbb{R}^{0|1}}(\Pi TM)$ *are equivalent as categories enriched over* $\mathrm{Ch}^{\mathbb{Z}/2}$.

Most of the identification we have already seen. The key input is the following Lemma.

Lemma 5.0.2. *Let* $E \to \Pi TM$ *be a super vector bundle. Then there is a super vector bundle* $E' \to M$ *and an isomorphism* $E \cong \pi^* E'$.

More precisely, the map $\eta_E : \pi^* z^* E \to E$ *is an isomorphism, where* $z : M \to \Pi TM$ *is the zero section.*

Proof. Choose a superconnection $\mathbb{\nabla}$ on E. Recall the canonical action of the multiplicative monoid $\mathbb{R}$ on ΠTM by dilation.

$$\begin{aligned} H : \mathbb{R} \times \Pi TM &\longrightarrow \Pi TM \\ (t, x^i, dx^i) &\longmapsto (x^i, t dx^i) \end{aligned}$$

Explicitly, we have $H^*\omega = t^{|\omega|}\omega$.

We consider the super parallel transport Φ_H along the family of (super)paths H from $(t_1, \theta_1) = (0, 0)$ to $(t_2, \theta_2) = (1, 0)$ which is an invertible bundle map

$$\Phi_H : H_0^* E \to H_1^* E\,.$$

This is equivalently a global isomorphism $H_0^* E \cong E$. Note that $H_0 = z \circ \pi$ is the composition of π with the zero section $z : M \to \Pi TM$. Hence $E \cong \pi^* z^* E$. □

For the relevant statement on super parallel transport, c.f. [Dum08, Theorem 3.5.].

Proof of the Proposition. Let $(E, \mathbb{\nabla})$ be a super vector bundle with superconnection over M. Recall that $\Gamma(\pi^* E) = \Omega(M; E)$. On such a differential form with values in E we can define $a^*_{\mathbb{\nabla}} : \Omega(M; E) \to \Omega(M; E)[\theta]$ by

$$a^*_{\mathbb{\nabla}}(\sigma) = \sigma + \theta \mathbb{\nabla}\sigma\,.$$

This defines a bundle map $a_{\mathbb{\nabla}} : \mathbb{R}^{0|1} \times \pi^* E \to \pi^* E$ since it indeed covers the base $\mathbb{R}^{0|1}$-action act on ΠTM given by $\mathrm{act}^*(\omega) = \omega + \theta d\omega$ and this is the basically content of the Leibniz rule:

$$\begin{aligned} a^*_{\mathbb{\nabla}}(\omega\sigma) &= \omega\sigma + \theta \mathbb{\nabla}(\omega\sigma) = \omega\sigma + \theta(d\omega\sigma + (-1)^{|\omega|}\omega \mathbb{\nabla}\sigma) \\ &= (\omega + \theta d\omega)(\sigma + \theta \mathbb{\nabla}\sigma) = (a^*\omega)(a^*_{\mathbb{\nabla}}\sigma)\,. \end{aligned}$$

$$\begin{array}{ccc} \mathbb{R}^{0|1} \times \pi^* E & \xrightarrow{a_{\mathbb{\nabla}}} & \pi^* E \\ \downarrow & & \downarrow \\ \mathbb{R}^{0|1} \times \Pi TM & \xrightarrow{a} & \Pi TM \end{array}$$

We can now define the integration functor:

$$\begin{aligned} \textstyle\int : \mathrm{sVect}^{\nabla}(M)_{\mathrm{flat}} &\longrightarrow \mathrm{sVect}^{\mathbb{R}^{0|1}}(\Pi TM) \\ (E, \mathbb{\nabla}) &\longmapsto (\pi^* E, a_{\mathbb{\nabla}}) \\ (\varphi : E_1 \to E_2) &\longmapsto (\varphi : \pi^* E_1 \to \pi^* E_2) \end{aligned}$$

We write φ instead of $\pi^*\varphi$ abusively since this is just the extension from sections of E to E-valued forms. The integration functor preserves the differentials. Write the action on E_i by $a_{E_i} = \mathrm{id} + \theta A_i$ to get:

$$D(\textstyle\int(\varphi)) = \varphi A_1 - (-1)^{|\varphi|} A_2\varphi = \varphi \circ \nabla_1 - (-1)^{|\varphi|}\nabla_2 \circ \varphi = \textstyle\int(D\varphi)\,.$$

We used only previous computations and $a_{\nabla_i} = \mathrm{id} + \theta\nabla_i = \mathrm{id} + \theta A_i$.

We now consider the zero section $z : M \to \Pi TM$. Let $E \to \Pi TM$ be a bundle with $\mathbb{R}^{0|1}$-action a_E. Then $z^*E \to M$ becomes a bundle with superconnection $\frac{\partial}{\partial\theta}a_E$. To see this we write $a_E = \mathrm{id} + \theta\frac{\partial}{\partial\theta}a_E$. Here, $\frac{\partial}{\partial\theta}a_E$ is an odd operator on $\Omega(M; z^*E) = \Gamma(E)$. As above, the compatibility with the $\mathbb{R}^{0|1}$-action on ΠTM is equivalent to the Leibniz rule for $\frac{\partial}{\partial\theta}a_E$. Hence, $\frac{\partial}{\partial\theta}a_E$ is a superconnection on z^*E. The functor that assigns the infinitesimal data is:

$$\begin{aligned}\mathrm{Inf} : \mathrm{sVect}^{\mathbb{R}^{0|1}}(\Pi TM) &\longrightarrow \mathrm{sVect}^{\nabla}(M)_{\mathrm{flat}}\\ (E, a_E) &\longmapsto (z^*E, \frac{\partial}{\partial\theta}a_E)\\ (\varphi : E_1 \to E_2) &\longmapsto (z^*\varphi : z^*E_1 \to z^*E_2)\end{aligned}$$

Note that $z^*\pi^*\varphi = \varphi$ and also that we identified $\pi^*\varphi = \varphi$ upon extension by forms. Again, this is compatible with the differential:

$$d(z^*\varphi) = \varphi \circ \frac{\partial}{\partial\theta}a_{E_1} - (-1)^{|\varphi|}\frac{\partial}{\partial\theta}a_{E_2} \circ \varphi = z^*d\varphi$$

We used that $a_{E_i} = \mathrm{id} + \theta\frac{\partial}{\partial\theta}a_{E_i}$.

We have the following two dg-functors:

$$\textstyle\int : \mathrm{sVect}^{\nabla}(M)_{\mathrm{flat}} \rightleftarrows \mathrm{sVect}^{\mathbb{R}^{0|1}}(\Pi TM) : \mathrm{Inf}$$

Clearly, $\mathrm{Inf} \circ \mathrm{OddTP} = \mathrm{id}$ since $\pi \circ z = \mathrm{id}_M$. This means OddTP is fully faithful. It is also essentially surjective: Any bundle on ΠTM is isomorphic to a pullback bundle.

Let $E \to \Pi TM$ be a bundle. We have seen before that the natural map $\eta_E : \pi^*z^*E \to E$ is an isomorphism of super vector bundles. Moreover, the following diagrams commute by definition of the maps:

$$\begin{array}{ccc}\mathbb{R}^{0|1} \times \pi^*z^*E & \xrightarrow{a_{\partial_\theta a_E}} & \pi^*z^*E\\ \downarrow{\scriptstyle \mathrm{id}\times\eta_E} & & \downarrow{\scriptstyle \eta_E}\\ \mathbb{R}^{0|1} \times E & \xrightarrow{a_E} & E\end{array} \qquad \begin{array}{ccc}\pi^*z^*E_1 & \xrightarrow{\pi^*z^*\varphi} & \pi^*z^*E_2\\ \downarrow{\scriptstyle \eta_{E_1}} & & \downarrow{\scriptstyle \eta_{E_2}}\\ E_1 & \xrightarrow{\varphi} & E_2\end{array}$$

That means, η is a (dg enriched) natural isomorphism $\textstyle\int \circ \mathrm{Inf} \to \mathrm{id}$. □

Theorem 5.0.3 (cite Camilo)**.** *The categories* $\mathsf{Flat}_{\mathbb{Z}/2}(M)$ *and* $\mathrm{Fun}(\Pi_\infty(M), \mathrm{Ch}^{\mathbb{Z}/2})$ *are equivalent as categories enriched over* $\mathrm{Ch}^{\mathbb{Z}/2}$*.*

Theorem 5.0.4 (cite Lurie)**.** *The* ∞*-categories* $\mathcal{S}\mathrm{hv}_\infty^{loc.const.}(M, \mathrm{Ch}^{\mathbb{Z}/2})$ *and* $\mathrm{Fun}(\Pi_\infty(M), \mathrm{Ch}^{\mathbb{Z}/2})$ *are equivalent.*

Recall that there is an equivalence of stable ∞-categories $\mathrm{Ch}^{\mathbb{Z}/2} \simeq \mathcal{D}(\mathbb{C}[u, u^{-1}]) \simeq \mathcal{M}\mathrm{od}(H\mathbb{C}[u, u^{-1}])$, where the generator u sits in degree $|u| = 2$.

Corollary 5.0.5. *The* ∞*-categories* $\mathsf{Flat}_{\mathbb{Z}/2}(M)$ *and* $\mathsf{sVect}^{\mathbb{R}^{0|1}}(\Pi TM)$ *are equivalent.*

5.1. **The complex case.**

Proposition 5.1.1. *Let $M \in \mathcal{M}\mathrm{fld}$. The dg-categories $\mathsf{Flat}_{\mathbb{Z}}(M)$ and $\mathsf{sVect}_{\mathbb{C}}^{\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})}(\Pi T_{\mathbb{C}}M)$ are equivalent.*

Proof. We use $\pi : \Pi T_{\mathbb{C}}M \to M$ for the projection and $z : M \to \Pi T_{\mathbb{C}}M$ for the zero section. Let $E \to M$ be a graded complex vector bundle with a flat $\mathbb{Z}$-graded connection D. Then, $\pi^*E \to \Pi T_{\mathbb{C}}M$ can be equipped with the action of $\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})$ given on section $\sigma \in \Gamma(\pi^*E) = \Omega(M;E)$ by

$$(z,\theta).\sigma = z^{|\sigma|}(\sigma + \theta D\sigma)\,.$$

Conversely, for any complex super vector bundle $E \to \Pi T_{\mathbb{C}}M$ with a $\mathrm{Diff}(\mathbb{R}^{0|1}_{\mathrm{cs}})$-action we have seen that $z^*E \to M$ carries a $\mathbb{Z}$-grading and a flat superconnection. The functors act on morphisms by pullback. The rest of the proof procedes as in the periodic case. □

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